Q6. Discuss the following terms; Facts, Measures, Dimensions

Facts represent the subject of the desired analysis; Facts are the important entity to be analyzed, a sale, a course, eg. “Serap took COMPE 341” . Facts most often identified via its dimension values.

Measures are numerical data being tracked. Measures represent the fact property that the

users want to study and optimize. A measure has two components:

- Numerical value: (sales price)
- Aggregation formula (SUM): used for aggregating/combining a number of measure
- values into one measure value determined by dimension value combination
- Dimensions are business parameters that define a transaction, time, space, products, Location

Example: Sale has the dimensions : Product , store, time

Q7. What is the 3 types of conceptual modeling of data warehouse?

Star Schema: Fact table in the middle connected to a set of dimension tables.

Snowflake Schema: Refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables

Fact constellations (Galaxy Schema): Multiple fact tables share dimension tables, viewed as a collection of stars.

Explanation: 3 Types of Schema Used in Data Warehouses

Star Schema only permits one central table and several de-normalized single-dimension tables joined to the table.

Characteristics of the Star Schema:

Star data warehouse schemas create a denormalized database that enables quick querying responses.

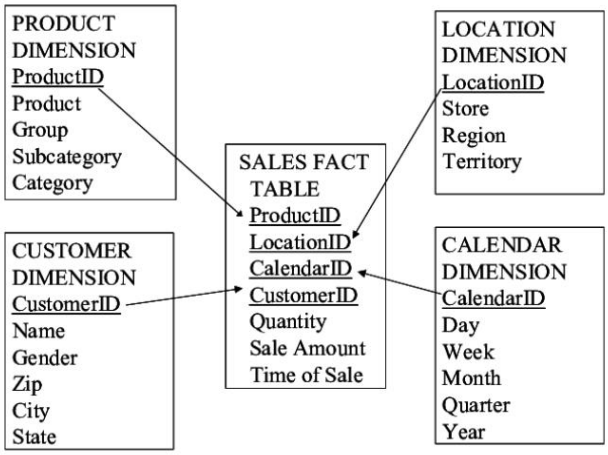
The primary key in the dimension table is joined to the fact table by the foreign key

Each dimension in the star schema maps to one-dimension table.

Dimension tables within a star scheme are not to be connected directly

Star schema creates de-normalized dimension tables.

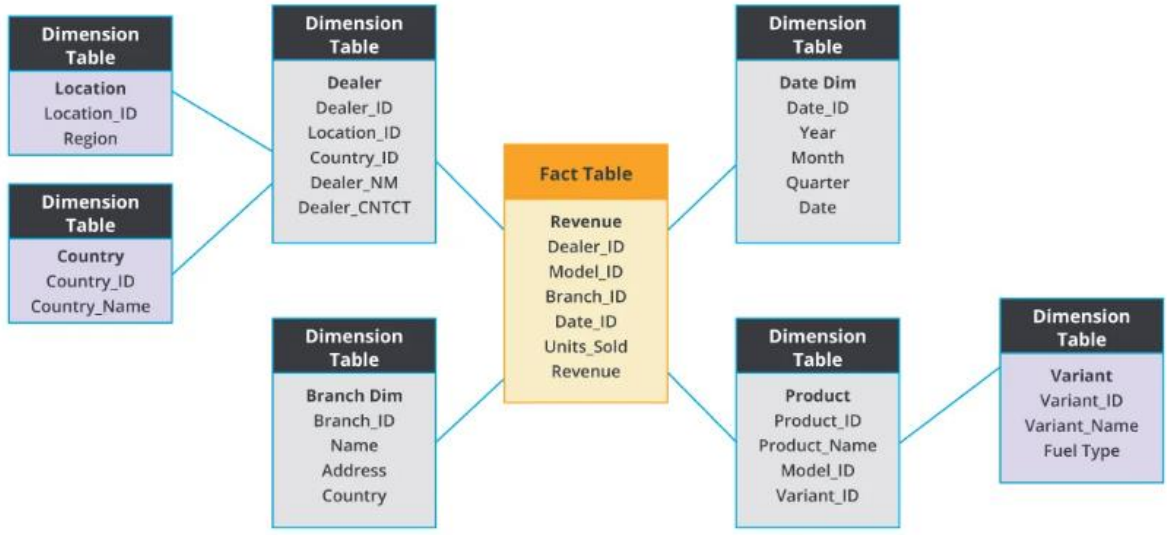
Example:



An Example Star Schema

Snowflake Schema

The Snowflake Schema is a data warehouse schema that encompasses a logical arrangement of dimension tables. This data warehouse schema builds on the star schema by adding additional sub-dimension tables that relate to first-order dimension tables joined to the fact table. Just like the relationship between the foreign key in the fact table and the primary key in the dimension table, with the snowflake schema approach, a primary key in a sub-dimension table will relate to a foreign key within the higher order dimension table. Snowflake schema creates normalized dimension tables – a database structuring strategy that organizes tables to reduce redundancy.

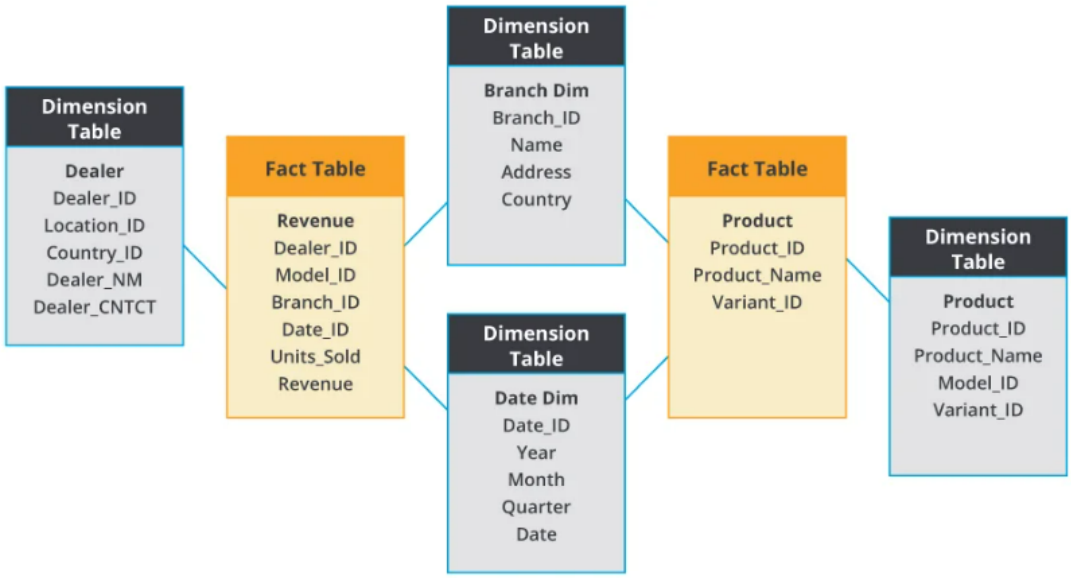


Example of Snowflake Schema

The Galaxy Schema (Fact Constellation Schema)

Acts as the next iteration of the data warehouse schema. Unlike the Star Schema and Snowflake Schema, the Galaxy Schema uses multiple fact tables connected with shared normalized dimension tables. Galaxy Schema can be thought of as star schema interlinked and completely normalized, avoiding any kind of redundancy or inconsistency of data.

Galaxy Schema is multidimensional acting as a strong design consideration for complex database systems.



Example of Galaxy Schema

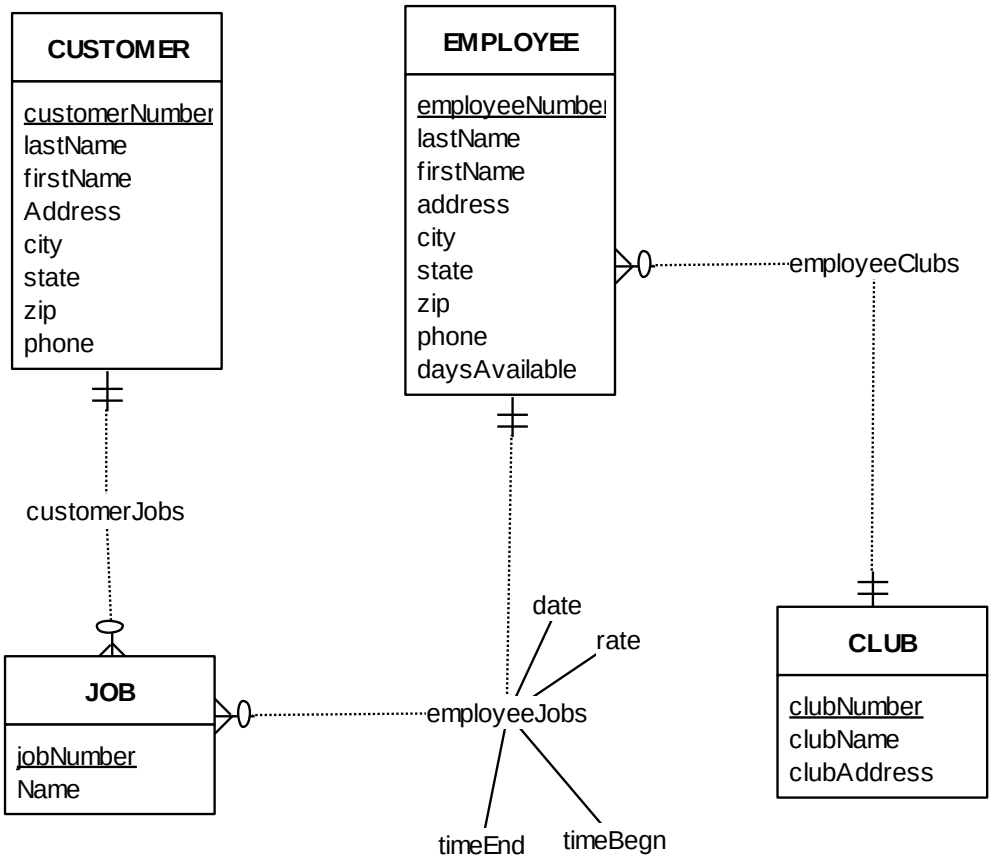
Reference: https://streamsets.com/blog/schemas-data-warehouses-star-galaxy-snowflake/

Q8. Creating a Data Warehouse

OU Babysitter Club

The OU Business Association is conducting a babysitter service as a fundraiser for different clubs in the college. When a customer is entered into the system, the OU Club coordinator gets name, address, and phone. The coordinator also records each babysitting job, the amount paid for it and the sitter assigned to the job. Each person may sign up to credit only one club and the system keeps the contact person and phone number for each participating club. The operational system conceptual design according to these requirements is provided below:

The management of the company wants a data warehouse for this system. He would like to be able to determine how much each customer was billed by week, month or year. How much each employee earned, also summed by time periods. He is interested in how much work is done on weekends, holidays or other special days. Develop a data warehouse to provide this information.



a) Define a fact according to this information
b) Define related dimensions according to the given conceptual model
c) Define related measures
d) Design the dimensional model by using STAR schema

Answers:

a) Define a fact according to this information - **JOB**

b) Define related dimensions according to the given conceptual model
Club, Customer, Date, Employee

c) Define related measures
Duration, Payment Amount

d) Design the dimensional model by using STAR schema

